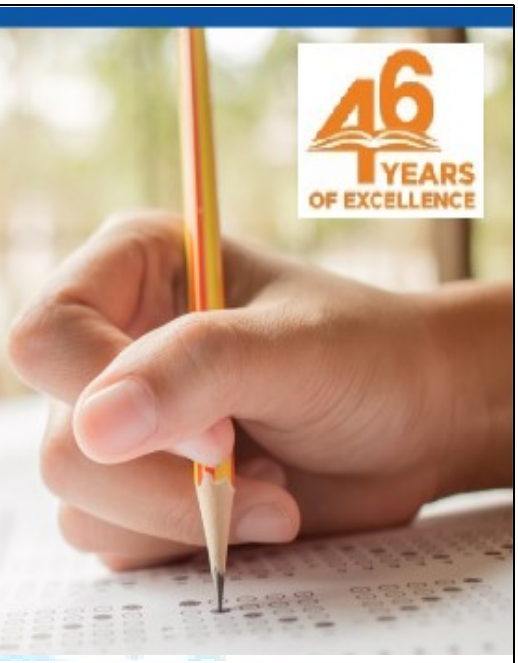
NARAYANA IIT ACADEMY INDIA

JEE ADVANCED GRAND TEST SERIES



46 YEARS OF EXCELLENCE

NARAYANA IIT - JEE ACADEMY - INDIA

Sec: OSR.IIT _*CO-SC(MODEL-A&APEX) **Date: 23-04-26**
Time: 3 Hrs **Max. Marks: 180**

23-04-26_OSR.IIT_*CO-SC(MODEL-A&APEX)_JEE ADV_GTA-8(P1)_SYLLABUS

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MATHEMATICS: **TOTAL SYLLABUS**

PHYSICS: **TOTAL SYLLABUS**

CHEMISTRY: **TOTAL SYLLABUS**

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TIME:3hrs **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 1 – 4)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 5 – 7)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 8 – 13)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Sec – IV (Q.N : 14 – 16)	Questions with **MATRIX MATCH**	+4	-1	3	12
Total				**16**	**60**

PHYSICS

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 17 – 20)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 21 – 23)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 24 – 29)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Sec – IV (Q.N : 30 – 32)	Questions with **MATRIX MATCH**	+4	-1	3	12
Total				**16**	**60**

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CHEMISTRY

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I (Q.N : 33 – 36)	Questions with **Single Correct** Options	+3	-1	4	12
Sec – II (Q.N : 37 – 39)	Questions with **Multiple Correct** Choice (partial marking scheme) (+1,0)	+4	-2	3	12
Sec – III (Q.N : 40 – 45)	Questions with Numerical Value Type (e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)	+4	0	6	24
Sec – IV (Q.N : 46 –48)	Questions with **MATRIX MATCH**	+4	-1	3	12
Total				**16**	**60**

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MATHEMATICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

1. Number of irrational roots of the equation $x^3-6x^2+12x-1=0$ is/are

A) zero B) 1 C) 2 D) 3

2. 3 students S_1, S_2, S_3 just after attempting the JEE Advance exam 2026 paper discuss the four events E_1,E_2,E_3,E_4 and agree on their probabilities in given below manner.

E_1 : No one scores above 75% marks

E_2 : S_3 scores above 75% marks

E_3 : S_3 scores above 75% marks given that at least two of them score above 75% marks

E_4 : Exactly one of them scores above 75% marks

$P(E_1)=\frac{1}{2},P(E_2)=\frac{1}{5},P(E_3)=\frac{1}{10},P(E_4)=\frac{1}{4}$

Find the probability that only S_3 score above 75% marks.

A) $\frac{1}{40}$ B) $\frac{3}{40}$ C) $\frac{7}{40}$ D) $\frac{9}{40}$

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3. The equation of the plane P_1 parallel to the line $L:\frac{x-1}{2}=\frac{y-0}{3}=\frac{z-2}{1}=\lambda$, which is at a distance of $\sqrt{6}$ units from the intersection point of L and plane $P:2x+3y+z-18=0$ and has maximum distance from $Q(5,4,-4)$ is $ax+by+cz+d=0$, the value of $\frac{bd}{ac}$ is :

A) $-\frac{3}{7}$ B) $\frac{3}{7}$ C) $\frac{15}{7}$ D) $-\frac{15}{7}$

4. Given that A and B are 3×3 invertible matrices such that $A^2B=BA$ and $A=P^{-1}$, then value of $P^3B^3A^3$ is

A) $A^{24}\ B^3$ B) $A^{21}\ B^3$ C) $A^{18}\ B^3$ D) $A^{15}\ B^3$

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SECTION - II (Maximum Marks: 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

5. If $f(x)=\|x|+|x-2[x]|-1|, x\in R$ and $g(x)=\frac{1}{2}\left(\left|x+\frac{1}{x}\right|+\left|x-\frac{1}{x}\right|\right), x\in R-\{0\}$, then which of the following is(are) true

(where [.] denotes greatest integer function)

A) $f\left(x-\frac{1}{2}\right)$ is an even function

B) $g(x)$ is non-differentiable at 2 points in its domain

C) $f(x)-g(x)=0$ has 2 solutions

D) $f(x)-g(x)=0$ has 3 solutions

6. Let f be a polynomial function defined only over non negative real numbers such that $f(x)f(y)+2=f(x)+f(y)+f(xy)$ and $f(x)$ is one-one function with $f(0)=1$ and $f'(1)=2$, then which of the following is/are true

A) The value of $f(5)$ is 26

B) If $h(x)=\min\left\{\frac{2}{f(x)}, x^2, |1-|x||\right\}$, where $x\geq 0$, then the number of points of non differentiability of $h(x)$ is 3

C) The value of $f(5)$ is 29

D) If $h(x)=\min\left\{\frac{2}{f(x)}, x^2, |1-|x||\right\}$, where $x\geq 0$, then the number of points of non differentiability of $h(x)$ is 2

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7. let R denote the set of all real numbers.

Let $S_1 = \{(x,y) \in \mathbb{R} \times \mathbb{R} : |x-2+i(y-3)|^2 + |x+i(y-5)|^2 = 8\}$

and $S_2 = \left\{(x,y) \in \mathbb{R} \times \mathbb{R} : \arg\left(\frac{x+iy}{x-2+iy}\right) = \frac{\pi}{2}\right\}$

where $i = \sqrt{-1}$,

represent two curves. If the points P and R lie on S_1 and the points Q and S lie on S_2 such that PQ and RS represent the limiting values of maximum and minimum values of the distances between S_1 and S_2 respectively, then

A) $PQ = 5 + \sqrt{2}$ B) $PQ = 5 + 2\sqrt{2}$

C) $RS = \sqrt{19 - 8\sqrt{2}}$ D) $RS = \sqrt{17} - \sqrt{2}$

SECTION-III (Maximum Marks: 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place**; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

8. Given that $\vec{a} = 2\hat{i} + \hat{j} + \hat{k}, \vec{b} = \hat{i} + 2\hat{j} - \hat{k}$ and $\vec{c}$ is such that $[\vec{a}\vec{b}\vec{c}] = 0, |\vec{c}| = \sqrt{2}$ and $\vec{c} \cdot \vec{a} = 0$. If $\vec{d}$ is a vector such that $(\vec{a} \times \vec{b}) \times \vec{d} = \vec{0}$ and $|\vec{d}| = \sqrt{3}$, then value of $|[\vec{b}\vec{c}\vec{d}]|$ is

9. Let n be a natural number such that $10 \le n \le 1000$. Let the mean deviation of n consecutive integers about the median be μ. If it is given that μ is a non-terminating decimal, then the number of distinct values of n possible for which μ is also a rational number is λ. The value of $\frac{\lambda}{82}$ is

10. Let $f : R \to R, f(x) = x^3 + 6x^2 - 15x + 3$. If $f'(c) = 0$ for some $c \in (a,b], a,b,c \in R$ such that $f(a) = f(b)$, then for the maximum possible value of $b, g(x) = bx^3 + ax^2 + cx + d$ is decreasing in the largest interval $[e,f]$, value of $3(e+f)$ is

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11. If the common tangent to the circle $x^2+y^2=16$ and ellipse $\frac{x^2}{64}+\frac{y^2}{4}=1$ in the first quadrant is a normal chord AB to the parabola $y^2=4ax, a>0$, normal at $A(x_1,y_1)$ and intersecting parabola again at $B(x_2,y_2)$. The value of $\frac{9}{\sqrt{5}}(x_1-y_2)$ is

12. If $\int\frac{(x^6-1-x)dx}{(6x^3+5x^2+30x^8\log_e x)^{7/8}}=\frac{4}{15}(f(x))^{\frac{1}{8}}+c, f(1)=11$

then the area bounded by $y=f(x), x=\frac{1}{2}, x=1$ and $y=11$ is $p+q\ln 2, p,q\in N$. The value of $p+q$ is :

13. Let $f(x)=\frac{1}{2}\left(\tan^{-1}\left(x-\frac{1}{2}\right)+\{3-x\}+\{5+x\}\right)$

and $I=\sum_{r=0}^{2022}\int_{\frac{1}{2025}}^{\frac{2}{2025}} f\left(x+\frac{r}{2025}\right)dx=\frac{a}{b}$

where $\gcd(a,b)=1, a,b\in I^+$. If $\left\{\frac{b}{a}\right\}=\frac{x}{a}$

then the value of x equals (where $\{.\}$ *represents fractional part of* x).

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SECTION-IV (Maximum Marks: 12)

This section contains **THREE (03)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

14. Let $A=\{1,2,3,4,\ldots\ldots,24,25,26\}$

A relation R is defined from set A to itself, then the number of

	List-I		List-II
P)	Relation R if $R=\{(a,b):R$ is asymmetric relation$\}$ are p^q (where p is prime number) then ($p+q$) equals	1)	328

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Q)	Relations R if $R=\{(a,b):R$ is reflexive but not symmetric$\}$ are $p^q\left(p^q-r\right)$, then $p+q+r$ is (where $p,q,r\in N$ and p is prime number)	2)	329
R)	Non-empty relations R if $R=\{(a,b):R$ is symmetric but not reflexive$\}$ are $p^q\left(p^r-1\right)-1$ then $p+q+r$ is (where p is prime number)	3)	330
S)	Non-empty relations R if $R=\{(a,b):R$ is neither symmetric nor reflexive $\}$ are $\left(p^s-1\right)p^q\left(p^q-1\right)$ then $p+q+s$ is (where p is prime number)	4)	353
		5)	356

A) $P\to1;Q\to1;R\to4;S\to4$ B) $P\to1;Q\to2;R\to3;S\to4$

C) $P\to3;Q\to2;R\to5;S\to5$ D) $P\to3;Q\to1;R\to4;S\to5$

15. Match each entry in List - I to the correct entry in List - II and choose the correct option.

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	List - I		List - II
P)	If $\dfrac{\left\|z-(5+3i)\right\|}{\left\|z-1-i\sqrt{3}\right\|}=\sin\left(\dfrac{\pi}{3}-\arg\left(z-1-i\sqrt{3}\right)\right)$ then locus of z is	1)	Line segment
Q)	If $5\left\|z-3\right\|=\left\|(4+3i)z+(4-3i)\bar{z}+2\right\|$ then locus of z is	2)	Pair of rays
R)	If $\left\|\left\|z-1\right\|-\left\|z-4+4i\right\|\right\|=5$ then locus of z is	3)	Circle
S)	If $\arg\left(\dfrac{z-20+25i}{z-2026i}\right)=0$, then locus of z is	4)	Parabola
		5)	Hyperbola

A) $P\to3;Q\to4;R\to5;S\to2$ B) $P\to3;Q\to4;R\to5;S\to3$

C) $P\to4;Q\to5;R\to2;S\to1$ D) $P\to4;Q\to5;R\to2;S\to2$

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16. If ${}^{a}C_{b}$ represents the binomial coefficient where ${}^{a}C_{b} = \frac{a!}{(a-b)!b!}$, where $a \in N, b \in W, a \geq b$ and ${}^{a}C_{b} = 0$, if $a < b$, then

Match each entry in List - I to the correct entry in List - II and choose the correct option.

	List - I		List - II
P)	If $\sum_{r=0}^{10} \frac{{}^{25}C_r \cdot {}^{10}C_{10-r}}{{}^{r+10}C_r} = \frac{{}^{p}C_q}{{}^{n}C_m}$, then the minimum value of $p+q+m+n$ is	1)	58
Q)	$\sum_{r=0}^{10} {}^{15}C_{10-r} \cdot {}^{25}C_r \cdot {}^{25-r}C_{10-r} = {}^{p}C_q \cdot {}^{n}C_m$, then the minimum value of $p+q+m+n$ is	2)	60
R)	$\sum_{r=0}^{16} (-1)^r {}^{25}C_r \cdot {}^{25-r}C_9 \cdot {}^{16}C_r = {}^{p}C_q \cdot {}^{n}C_m$, then the minimum value of $p+q+m+n$ is	3)	85
S)	$\sum_{r=0}^{18} (-1)^r \cdot {}^{28}C_r \cdot {}^{r}C_{10} \cdot {}^{18}C_r = {}^{p}C_q \cdot {}^{n}C_m$, then the minimum value of $p+q+m+n$ is	4)	70
		5)	110

A) $P \to 5; Q \to 4; R \to 3; S \to 2$ B) $P \to 5; Q \to 4; R \to 1; S \to 2$

C) $P \to 5; Q \to 4; R \to 1; S \to 3$ D) $P \to 3; Q \to 4; R \to 1; S \to 2$

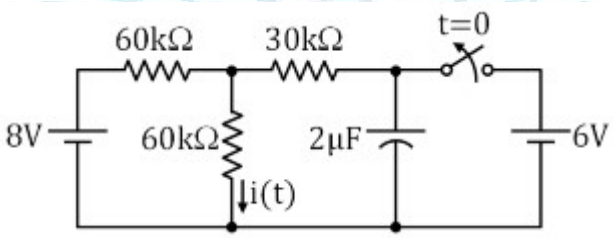
For More Join: @IITJEE_Advanced

PHYSICS **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

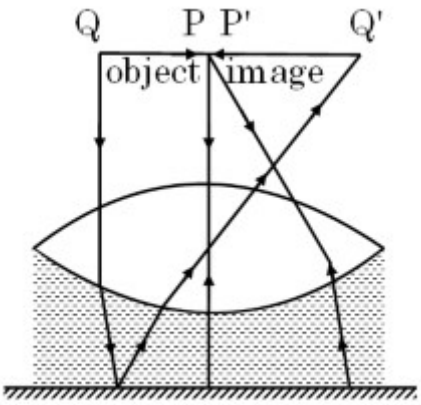
17. The circuit in figure is at steady state before the switch opens at $t=0$. Find the current $i(t)$ for $t>0$.



A) $\left(66.7+16.7e^{-\left(\frac{t}{120\ ms}\right)}\right)\mu A$ B) $\left(66.7-16.7e^{-\left(\frac{t}{60\ ms}\right)}\right)\mu A$

C) $\left(100-16.7e^{-\left(\frac{t}{120\ ms}\right)}\right)\mu A$ D) $\left(33.3+16.7e^{-\left(\frac{t}{120\ ms}\right)}\right)\mu A$

18. The figure shows an equiconvex lens (of refractive index 1.50) in contact with a liquid layer on top of a plane mirror. A small needle PQ with its tip on the principal axis is moved along the axis until its inverted image is found at the position of the needle. The distance of the needle from the lens is measured to be 45.0 cm . The liquid is removed and the experiment is repeated. The new distance is measured to be 30.0 cm . The refractive index of the liquid is

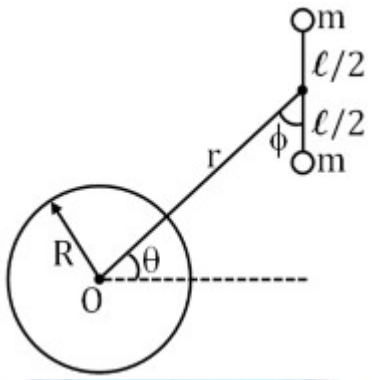


A) $\frac{5}{4}$ B) $\frac{9}{8}$ C) $\frac{4}{3}$ D) $\frac{6}{5}$

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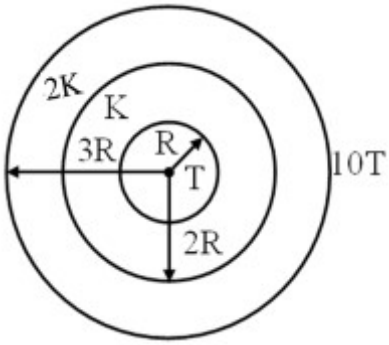
19. Two particles, each of mass m , are connected by a massless rod of length ℓ . The center of mass of this system is in a circular orbit of radius r about the earth. Calculate the angular frequency of small oscillations of the coordinate ϕ in the plane of the orbit.

Assume that $\ell \ll r$ and $|\phi| \ll 1$. $\left(g_0 = \frac{GM}{R^2}\right)$



A) $\sqrt{\frac{3\ g_0R^2}{r^3}}$ B) $\sqrt{\frac{g_0R^2}{r^3}}$ C) $\sqrt{\frac{g_0R}{r^2}}$ D) $\sqrt{\frac{3\ g_0R^3}{r^4}}$

20. A composite spherical shell is made up of two materials having thermal conductivities K and 2 K respectively as shown in the figure. The temperature at the inner most surface is maintained at T whereas the temperature at the outer most surface is maintained at 10 T . The net rate of heat flow from the outermost surface to the innermost surface of the shell will be :- (Neglect any radiation)



A) $63\pi RKT$ B) $72\pi RKT$ C) $\frac{432}{7}\pi RKT$ D) $\frac{147}{7}\pi RKT$

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SECTION - II (Maximum Marks: 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

21. A particle of mass m is attracted to a fixed point O by a force that varies directly with its distance r , that is, $F_r = -kr$. It undergoes orbital motion according to the initial conditions $r(0) = \ell, \dot{r}(0) = \ell\sqrt{\frac{k}{m}}, \theta(0) = 0$, $\dot{\theta}(0) = \sqrt{\frac{k}{m}}$. (where θ represents phase of the motion at any instant)

A) The value of r_{max} is approximately 1.6ℓ .

B) The value of r_{min} is approximately 0.6ℓ .

C) The time period of the orbit is $2\pi\sqrt{\frac{m}{k}}$.

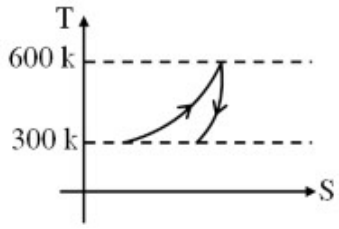
D) Angular momentum about O remains conserved.

22. $1.2\ m^3$ of air is heated reversibly at constant pressure from 300 K to 600 K , and is then cooled reversibly at constant volume back to initial temperature. If the initial pressure is 1 bar, (Take : $c_p = 1.005\ kJ/kgK$ and $R = 0.287\ kJ/kgK$, $\ell n2 = 0.7$)

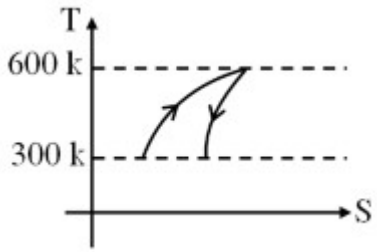
A) The net heat flow in the process is 120 kJ .

B) The net change in entropy is $0.28\ kJ/k$.

C) The T-S curve is given by

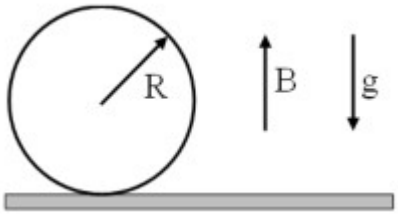


D) The T-S curve is given by



For More Join: @IITJEE_Advanced

23. A uniform thin-walled cylindrical tube of radius R , length L , and mass m is made of an insulating material and carries a total charge q with uniform surface charge density. The tube rolls on a horizontal table, and the entire system is placed in a vertical, uniform magnetic field of induction B and in a uniform gravitational field of strength g (see figure). The coefficient of friction between the tube and the table is μ. Determine the maximum value of the velocity v_{max} for which the tube rolls without slipping on the table, while being in contact with the table at more than one point.



A) If $L < 3\mu R, v_{max}$ is equal to $\frac{mgL}{3qBR}$

B) If $L < 3\mu R, v_{max}$ is equal to $\frac{mgL}{qBR}$

C) If $L \geq 3\mu R, v_{max}$ is equal to $\frac{\mu mg}{qB}$

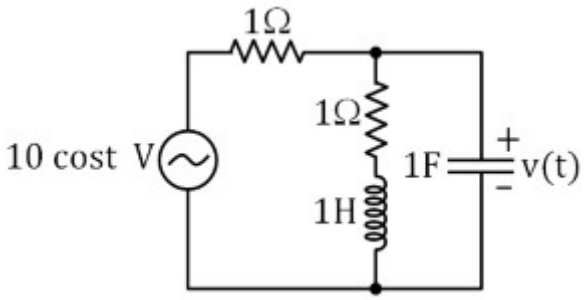
D) If $L \geq 3\mu R, v_{max}$ is equal to $\frac{2\mu mg}{qB}$

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SECTION-III (Maximum Marks: 24)

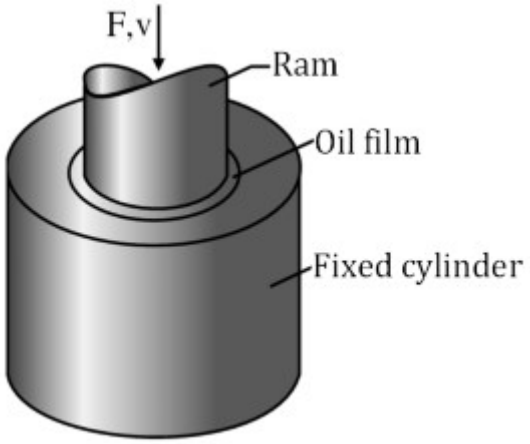
This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

24. The maximum value of $V(t)$ (in V) in the RLC circuit of figure is :-



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25. A hydraulic lift of the type commonly used for greasing automobiles consists of a $280.00-\text{mm}$-diameter ram which slides in a $280.18-\text{mm}$-diameter cylinder (similar to figure), the annular space being filled with oil having a kinematic viscosity of $0.00042\ \text{m}^2/\text{s}$ and specific gravity of 0.86. If the rate of travel of the ram is $0.22\ \text{m}/\text{s}$, find the frictional resistance (in N) when 2 m of the ram is engaged in the cylinder.



26. The following deuterium reactions and corresponding reaction energies Q are found to occur.

$^{14}\text{N}(\text{d},p)^{15}\text{N}, \quad Q=8.53\text{MeV}$

$^{15}\text{N}(\text{d},\alpha)^{13}\text{C}, \quad Q=7.58\text{MeV}$

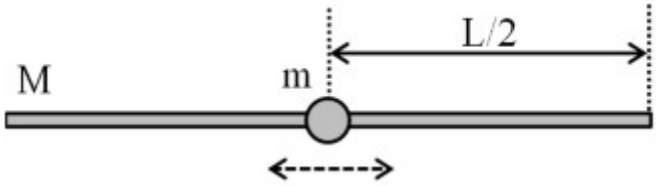
$^{13}\text{C}(\text{d},\alpha)^{11}\text{B}, \quad Q=5.16\text{MeV}$

What is the Q value (in MeV) of the reaction $^{11}\text{B}(\alpha,n)^{14}\text{N}$?

$^4_2\text{He}=4.0026\ \text{m}_\text{u}, {}^2_1\text{H}=2.0140\ \text{m}_\text{u}, {}^1_1\text{H}=1.0078\ \text{m}_\text{u}, \text{n}=1.0087\ \text{m}_\text{u}$.

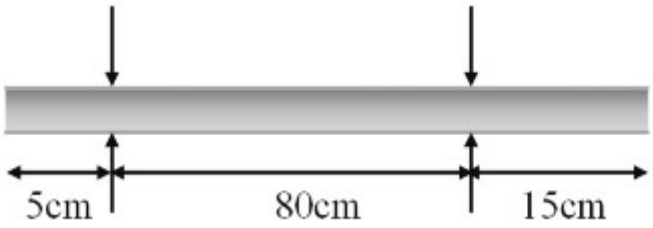
(One twelfth of the mass of $^{12}\text{C}=931\text{MeV}/\text{c}^2$.)

27. A homogeneous, smooth, thin, sufficiently long rod of mass M and length L is in a state of weightlessness in deep space (figure). A small bead is threaded onto the rod; the bead has mass m, which is much smaller than the mass of the rod $(\text{m}\ll\text{M})$. If the period T of small oscillations of the bead near the center of the rod is given by $\alpha\sqrt{\dfrac{\text{L}^3}{\text{GM}}}$. The value of α is. (The gravitational constant is G.)



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28. A metal rod of length 1 m is clamped at two points as shown in the figure. Distance of the clamp from the two ends are 5 cm and 15 cm respectively. Find the minimum natural longitudinal oscillation (in KHz) of the rod. Given that Young's modulus of elasticity and density of aluminium are $Y = 1.6 \times 10^{11} Nm^{-2}$ and $\rho = 2500 kgm^{-3}$ respectively.



29. Narrow, parallel, glowing gas-filled tubes in a variety of colors form block letters to spell out the name of a night club. Adjacent tubes are all 2.44 cm apart. The tubes forming one letter are filled with neon and radiate predominantly red light with a wavelength of 650 nm . For another letter, the tubes emit predominantly blue light at 390 nm . The pupil of a dark-adapted viewer's eye is 5.20 mm in diameter. If she is in a certain range of distance, the viewer can resolve the separate tubes of one color but not the other. The viewer's distance must be in a range, for her to resolve the tubes of only one of these two colors is given by $\alpha < d < \beta$. The value of $(\beta - \alpha)$ (in m) is :-

SECTION-IV (Maximum Marks: 12)

This section contains **THREE (03)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

30. An X-ray tube operates with a metal target. Assume ideal conditions unless stated. Match each item in List-I with the correct option from List-II. ($hc = 1240 eVnm$)

List-I (Situations/Observations)		**List-II** (Correct Consequences/Statements)	
P)	The accelerating potential of an X-ray tube is increased from V to 2 V	1)	The minimum wavelength of X-rays decreases by a factor of 2
Q)	The target material of an X-ray tube is changed from Cu to Mo, while keeping the accelerating voltage the same and sufficiently high.	2)	The positions of characteristic Lines change, but the Minimum wavelength remains unchanged.

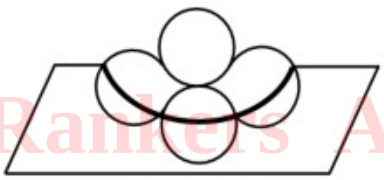
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R)	In an X-ray spectrum, a sharp line is observed whose frequency varies linearly with $(Z-1)^2$	3)	The observed line corresponds to a characteristic X-ray and obeys Moseley's law.
S)	The filament current of the X-ray tube is increased While keeping the Accelerating voltage and Target unchanged.	4)	The intensity of both continuous and characteristic spectra increases, but their Wavelength remain unchanged.
		5)	The observed line corresponds To continuous X-ray.

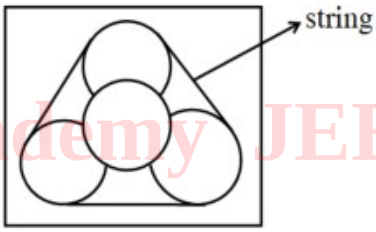
A) $P \to 2; Q \to 4; R \to 5; S \to 3$ B) $P \to 3; Q \to 2; R \to 3; S \to 1$

C) $P \to 1; Q \to 2; R \to 5; S \to 3$ D) $P \to 1; Q \to 2; R \to 3; S \to 4$

31. Three identical smooth spheres of mass 3 kg each are kept on horizontal smooth floor and one held together by a light string wrapped around them, at the level of their centres. A fourth identical smooth sphere is kept symmetrically on the top of the three, forming a pyramid like structure ($g = 10\ m/s^2$).



Side view



Top view

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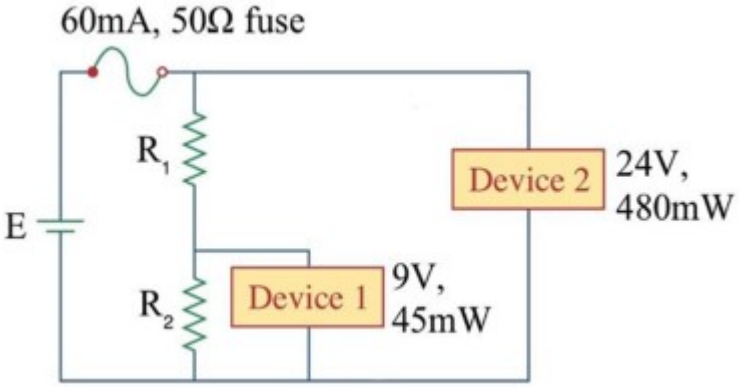
	List -I		List-II (All values in SI units)
P)	The minimum tension in the string to hold the system in equilibrium.	1)	10
Q)	Normal force on each sphere by the floor	2)	$10\sqrt{\frac{3}{2}}$
R)	Normal force by any bottom sphere on the sphere at the top	3)	$\frac{10}{\sqrt{6}}$
S)	Upon cutting the string, initial acceleration of the sphere at the top	4)	$\frac{10}{7}$
		5)	40

A) $P \to 3; Q \to 5; R \to 2; S \to 4$ B) $P \to 2; Q \to 3; R \to 5; S \to 1$

C) $P \to 3; Q \to 1; R \to 2; S \to 1$ D) $P \to 4; Q \to 2; R \to 5; S \to 3$

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32. The device-1 & 2 as well as the fuse are operating at the rated powers in the shown circuit.



	List-I		List-II (All values in SI units)
P)	The value of R_1	1)	257.1
Q)	The value of R_2	2)	32.4
R)	The value of battery EMF E	3)	52.7
S)	The percentage efficiency of the circuit (i.e., ratio of the total power consumed by the two devices & the total power)	4)	27
		5)	375

A) $P \to 5; Q \to 1; R \to 4; S \to 2$
B) $P \to 2; Q \to 4; R \to 1; S \to 3$
C) $P \to 5; Q \to 2; R \to 4; S \to 2$
D) $P \to 5; Q \to 2; R \to 3; S \to 2$

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CHEMISTRY **MAX.MARKS: 60**

SECTION–I (Maximum Marks: 12)

This section contains **FOUR (04)** questions.
Each question has FOUR options for correct answer(s). **ONLY ONE** of these four option is the correct answer.
For each question, choose the correct option corresponding to the correct answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks : +3 If only the correct option is chosen.
Zero Marks: 0 If none of the option is chosen.(i.e the question is un answered)
Negative Marks: -1 In all other cases.

33. Following steps are generally involved in extraction of metals from their ores
(I) Roasting
(II) Froth flotation in which NaCN is used as depressant
(III) Magnetic separation
(IV) Electrorefining in which pure Al is taken as cathode
(V) Smelting
(VI) Distillation

Select the correct option regarding above steps involved in extraction of Zn from Zinc blende

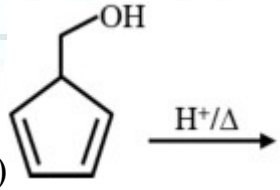
A) (I), (II), (III), (V), (VI)
B) (I), (IV), (V), (VI)
C) (I), (II), (IV), (V)
D) (III), (V), (VI)

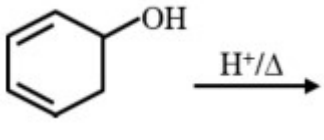
34. An orbital of H -atom is represented as :

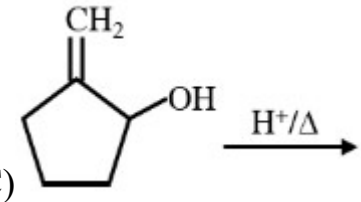
$$\psi_{n,\ell,m_\ell}(r,\theta,\phi)=\frac{\sqrt{2}}{81\sqrt{\pi}a_0^{3/2}}\left[6-\frac{r}{a_0}\right]e^{\left(-\frac{r}{3a_0}\right)}\sin\theta\cdot\sin\phi,\ \text{where}\ a_0=0\cdot529\text{Å}$$

Then select the incorrect statement based on above information
A) Number of radial nodes in this orbital is 1
B) It represents p_y orbital
C) orbital angular momentum of the e^- in orbital is $\sqrt{2}\hbar$
D) Probability of finding electron is maximum at distance 3.174Å from nucleus

35. Phenol upon treatment with Zn dust produces an aromatic hydrocarbon ' X '. In which of the following reaction(s) aromatic hydrocarbon ' X ' can not be synthesized ?

A) 

B) 

C) 

D) $HOOC-(CH_2)_6-COOH \xrightarrow[\text{(ii) } Cr_2O_3,\ 773K,\ 10\text{-}20\ atm]{\text{(i) Sodalime}/\Delta}$

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36. Consider given hydrides of nitrogen family and select correct statement.

NH_3, PH_3, AsH_3 and SbH_3

A) Only NH_3 have higher thermal stability and more basic as compared to PH_3.

B) Only AsH_3 & SbH_3 have higher boiling point as compared to NH_3.

C) Only SbH_3 have higher melting point as compared to NH_3.

D) Only NH_3 and PH_3 have higher bond angle than 100°.

SECTION - II (Maximum Marks: 12)

This section contains **THREE (03)** questions.
Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
For each question, choose the correct option(s) to answer the question.
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.
Partial Marks: +3 If all the four options are correct but ONLY three options are chosen.
Partial Marks: +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.
Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).
Negative Marks: -2 In all other cases.

37. Cells of a bacteria divide every 10 minutes under favourable conditions. A cell (considering cylindrical rod of length 2μ m & 1μ m in diameter) of this bacteria is placed on top of a nutrient rich soup taken in a cell-culture dish with a radius of R cm (top surface area $= 1.732 \times 10^{-2}$ m^2).

If **t** minutes are required to develop the colony of **N** number of this bacteria to cover the entire surface of dish considering cells are spread in single layer & bacteria stay vertically and closely packed as possible. Then select the correct statement(s)

$\left(\log 2 = 0.3, \log 3 = 0 \cdot 48, \sqrt{3} = 1.732, \sqrt{2} = 1.414\right)$

A) **N** is approximately 5×10^9

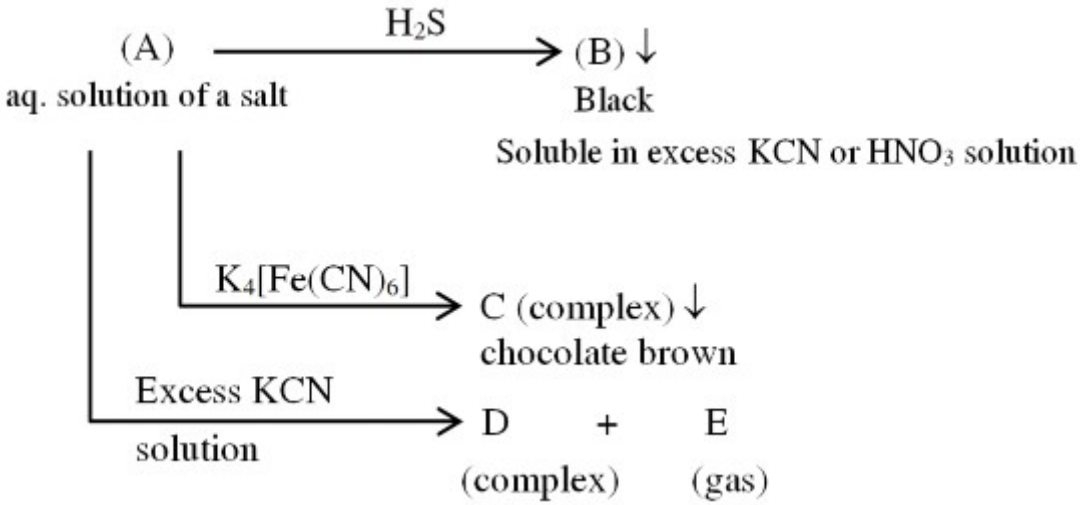
B) **t** is approximately 103.33 minutes

C) N is approximately 2.22×10^5

D) **t** is approximately 323.33 minutes

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38.



Select correct statement(s) based on above information

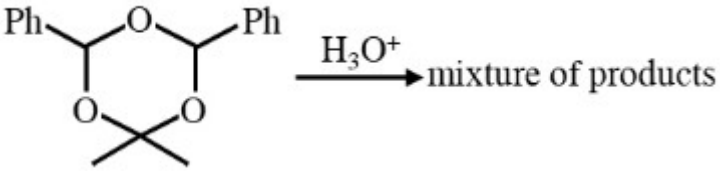
A) Hybridisation of central metal ion in (D) is sp^3

B) Gas (E) disproportionate when it is passed through aq. NaOH

C) (B) is insoluble in aqua-regia.

D) Compound (D) produce precipitate of metal sulphide with H_2S gas.

39. Consider following reaction



Which of the following statement(s) is/are correct?

A) One of the product can give positive iodoform test

B) One of the product can give positive Tollens test

C) One of the product can give negative Fehlings test

D) Products upon reaction with NH_2OH can form 4 different types of oximes

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SECTION-III (Maximum Marks: 24)

This section contains **SIX (06)** questions. The answer to each question is a **NUMERICAL VALUE**
For each question, enter the correct numerical value (in decimal notation, truncated/rounded off to the **second decimal place;** e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30) designated to enter the answer.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 If ONLY the correct numerical value is entered as answer.
Zero Marks: 0 In all other cases.

40. If 20% of Mn^{2+} ion in $1\times10^{-3}M$ solution of $MnSO_4$ (aq.) in acidic medium maintained at $pH = x$ is oxidised after a long exposure to air (20% oxygen by volume) at 1 bar &$25^{\circ}C$, then find 'x ' based on following information.

	E° at $25^{\circ}C$
$MnO_2 + 4H^+(aq.) + 2e^- \rightarrow Mn^{+2}(aq.) + 2H_2O;$	1.23 V
$O_2(g) + 4H^+(aq.) + 4e^- \rightarrow 2H_2O(\ell);$	1.23 V

$\left(\log2 = 0.3, \log3 = 0.48, \log5 = 0.7, \frac{2.303RT}{F} \text{ at } 25^{\circ}C = 0.059 \text{ V}\right)$

41. Find the pOH of resultant solution obtained at $25^{\circ}C$ when $0.2M, 500ml AcOH, 0.1M, 500ml$ barium acetate and $0.1M, 500ml NaOH$ are added into 500 ml of water.

Given that the volume is conservative.

(pK_a of $AcOH = 5, pK_w = 14$)

$(\log2 = 0.3, \log3 = 0.48, \log5 = 0.7)$

42. When 100 ml of saturated aqueous solution of cupric iodate (negligible amount of solid in contact with solution) is treated with excess KI (aq.) solution in acidic medium, the liberated I_2 requires 100 ml of 0.26 N $Na_2S_2O_3$ (aq.) solution in this iodometric titration. If K_{sp} of cupric iodate is represented as $x\times10^{y}$ (in scientific notation), then find ' x '.

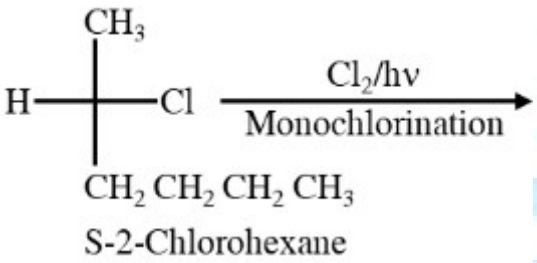
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43. If

$X =$ total number of isomeric alkynes (including stereoisomers) with molecular formula C_6H_{10} that can produce CH_4 upon treatment with CH_3MgBr

and

$Y =$ total number of achiral dichloro products formed in following reaction without considering free radical rearrangement.



S-2-Chlorohexane

Then value of $(X+Y)$ is

44. What is the percentage of chlorine in the monomer of Neoprene?

(Given : atomic masses of $Cl \Rightarrow 35.5, H \Rightarrow 1, C \Rightarrow 12$)

45. Find the sum of number of oxygen atoms in :-

(i) Pyrosulphurous acid

(ii) Solid Nitronium nitrate

(iii) Trimetaphosphoric acid

(iv) Periodic acid

(v) Tris(acetylacetonato)iron(III)

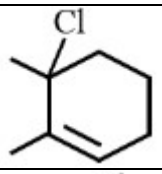
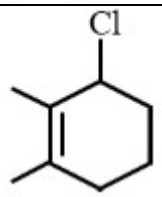
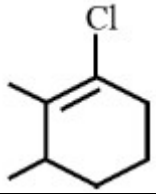
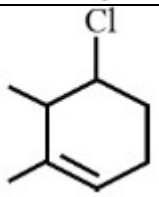
(vi) $EDTA^{4-}$

SECTION-IV (Maximum Marks: 12)

This section contains **THREE (03)** Matching List Sets. Each set has ONE Multiple Choice Question
Each set has TWO lists: List-I and List-II. List-I has Four entries (P), (Q), (R) and (S) and
List-II has Five entries (1), (2), (3), (4) and (5).
FOUR options are given in each Multiple Choice Question based on List-I and List-II and ONLY ONE of these four options satisfies the condition asked in the Multiple Choice Question.
Answer to each question will be evaluated according to the following marking scheme:
Full Marks: +4 ONLYif the option corresponding to the correct combination is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks: – 1 in all other cases.

46. List-I contains some compounds. List-II contains some numbers, each of which basically corresponds to sum of locants of substituents when IUPAC name of any compound in List-I is written. Match each entry in List-I with appropriate entry in List-II and choose correct option?

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LIST-I		LIST-II	
P)		1)	5
Q)		2)	6
R)		3)	11
S)		4)	12
		5)	13

A) $P \rightarrow 3; Q \rightarrow 2; R \rightarrow 4; S \rightarrow 4$ B) $P \rightarrow 4; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 3$

C) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 1; S \rightarrow 5$ D) $P \rightarrow 5; Q \rightarrow 2; R \rightarrow 2; S \rightarrow 4$

47. Match the carbohydrates given in List-I with type(s) of linkage(s) present in their structure given in List-II and choose the correct option

LIST-I		LIST-II	
P)	Amylose	1)	$\alpha-1,4$
Q)	Amylopectin	2)	$\alpha-1,6$
R)	Cellulose	3)	$\beta-1,4$
S)	Lactose	4)	$\beta-1,6$
		5)	$\alpha 1, \beta 2$

A) $P \rightarrow 2; Q \rightarrow 4; R \rightarrow 5; S \rightarrow 1,3$ B) $P \rightarrow 1,2; Q \rightarrow 1; R \rightarrow 3; S \rightarrow 5$

C) $P \rightarrow 1; Q \rightarrow 1,2; R \rightarrow 3; S \rightarrow 3$ D) $P \rightarrow 3; Q \rightarrow 5; R \rightarrow 1; S \rightarrow 4$

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48. Match the compounds in List - I with the appropriate properties in List - II and choose the correct option.

LIST-I (Complex compound)		LIST-II (Properties)	
P)	$\left[Ni(en)_2(NH_3)_2\right]Cl_2$	1)	inner orbital complex
Q)	$\left[Ni(en)_3\right]Cl_2$	2)	$\mu_s = 0$
R)	$\left[Cr(NH_3)_6\right]Cl_2$	3)	Show optical isomerism
S)	$\left[Fe(en)_2Cl_2\right]Cl$	4)	Complex has geometrical isomers
		5)	Aqueous solution (if possible) has significant electrical conductivity (more than water)

A) $P \rightarrow 1,2,3,4,5; Q \rightarrow 1,2,3,5; R \rightarrow 1,2,5; S \rightarrow 2,3,4,5$

B) $P \rightarrow 3,4,5; Q \rightarrow 3,5; R \rightarrow 5; S \rightarrow 1,3,4,5$

C) $P \rightarrow 3,4,5; Q \rightarrow 4,5; R \rightarrow 2,5; S \rightarrow 1,3,4,5$

D) $P \rightarrow 2,3,4,5; Q \rightarrow 2,3,5; R \rightarrow 5; S \rightarrow 3,4,5$

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